Quantitative Data Analysis and Reporting

What do we want to know? What data would answer it? Which method fits? What can the output support?

English lesson handout. Open the interactive lesson for everyday story examples. Specialist method explanations are linked from the map for independent exploration.

Open the interactive lesson

Learning aims

By the end of the lesson, you should be able to:

1. Report a sample clearly with appropriate descriptive statistics, units and denominators.

2. Explain common comparison, association, regression and reliability methods, including mediation and moderation, and read their main results.

3. Use the method map to explore further analyses when your research question requires them.

4. Consult the glossary and use JARS-Quant to check how analyses and results are reported.

Descriptive statistics

Who or what is in the sample, and what do the observed values look like?

What this method does

Describes the observed sample with frequencies, typical values, spread and distribution shape.

Required data and design

Scores or categories, with units, valid sample sizes and denominators.

Outputs to read

n, percentages, M and SD or median and IQR; the observed distribution.

What to report

1. Give the valid n and missing values for each variable; identify units and the score scale.

2. For categories, report counts and percentages with their denominator.

3. For continuous data, report centre and spread together: M and SD, or median and IQR where appropriate.

4. Use a labelled table or distribution plot. Describe the pattern without turning a sample summary into a population conclusion.

Research example: how the result is reported

Verbatim excerpt · Original English

> It shows that participants engaged more in RIA (M = 3.20, SD = 0.71) than PIA (M = 2.48, SD = 0.77).

Lee & Drajati (2019). Affective variables and informal digital learning of English: Keys to willingness to communicate in a second language
Results · PDF page 5 (checked copy; counted from its first page)

Reading the report

The authors name both activities and give a mean (M) and standard deviation (SD) for each. SD describes variation in scores; this sentence alone does not test the difference.

Basic t-tests: independent or paired?

Are two group means different, or is the average paired change different from zero?

What this method does

An independent t-test compares the means of two separate groups. A paired t-test analyses the within-pair differences, such as the same learners before and after a lesson.

Required data and design

A continuous outcome in two independent groups, or correctly paired observations.

Outputs to read

Group summaries, mean difference, 95% CI, t, df, p and a named effect size.

What to report

1. State whether the design is independent or paired, and identify the groups or occasions.

2. Report n, M and SD for each group or occasion; for paired data, also make the change and its direction clear.

3. Report the mean difference and its 95% confidence interval, t, degrees of freedom and p.

4. Report a named effect size, including the version used for paired data. Interpret magnitude and uncertainty alongside the test result.

Worked teaching calculation

For paired changes 2, 4, 4, 6: mean change = 4; SD = 1.63; SE = 1.63/√4 = 0.816; t = 4/0.816 = 4.90, df = 3.

Teaching values for following the calculation; these are not findings from the cited study.

Interpret the result

The paired test evaluates the average change, preserving the pairing.

Check the claim and reporting

Independent and paired tests have different standard errors. Welch handles unequal group variances.

Research example: how the result is reported

Verbatim excerpt · Original English

> The results of paired-samples t-tests showed that after a 10-month study abroad there was a significant improvement in all three measures with varying effect sizes

Kahng & Otonya (2021). What Elicited Imitation Can Show Us: Quantitative and Qualitative Analyses of Longitudinal Changes
Results · PDF page 5 (checked copy; counted from its first page)

Source table extract · Selected values, reformatted

Table 1 · EI Performance of Pretest and Posttest

Measure	Pretest M (SE)	Posttest M (SE)	95% CI	p	d
Accuracy (0-100)	71.74 (3.99)	83.47 (2.02)	[−18.28, −5.19]	.001	0.98
Meaning (1-5)	3.47 (0.14)	4.22 (0.11)	[−0.94, −0.56]	< .0001	1.79
Formulaic language (0-58)	39.50 (2.43)	45.67 (1.66)	[−10.66, −1.92]	.007	0.64

Source values transcribed into the lesson layout. The intervals correspond to pretest minus posttest differences. The original table does not supply t or degrees of freedom.

Kahng & Otonya (2021) · PDF page 5

Reading the report

The prose states the direction; Table 1 supplies means, standard errors, intervals, p and effect sizes. For accuracy, the posttest mean is higher and the pre-minus-post interval is entirely negative. SE is not SD. A fuller paired-test report would also identify t and degrees of freedom; they are absent from this table.

Correlation: direction and strength

When one variable is higher, does the other tend to be higher or lower?

What this method does

Pearson’s r describes the direction and strength of a linear relationship between two continuous variables. Begin with a scatterplot so the coefficient can be read in context.

Required data and design

Pairs of continuous scores from the same cases; check the scatterplot and observation structure.

Outputs to read

Complete-pair n, Pearson’s r, its interval and p when testing; a scatterplot.

What to report

1. Name the two variables and the correlation used; here the focus is Pearson’s r.

2. Report the number of complete pairs, r, its confidence interval and p when inferential testing is used.

3. Explain the sign and size in the study context, and inspect linearity and unusual observations.

4. A correlation does not establish causation. A value near zero rules out neither a curved relationship nor all forms of association.

Worked teaching calculation

Centred cross-products $\Sigma xy = 8$, $\Sigma x^2 = 10$, $\Sigma y^2 = 10$ give $r = 8/\sqrt{(10 \times 10)} = .80$.

Teaching values for following the calculation; these are not findings from the cited study.

Interpret the result

The toy association is positive; the plot determines whether a single coefficient is adequate.

Check the claim and reporting

Correlation is not causation. Pearson and Spearman target different features.

Research example: how the result is reported

Verbatim excerpt · Original English

> In particular, EIL perceptions had higher correlations with TOEIC scores (r = .56; p < .01) than speaking (r = .40; p < .01) and PVLT (r = .40; p < .01) (Table 2).

Lee (2020). The role of informal digital learning of English and a high-stakes English test on perceptions of English as an international language
Results · PDF page 6 (checked copy; counted from its first page)

Reading the report

Each r gives the strength and direction of a bivariate association. Different r values do not by themselves establish a statistically significant difference between correlations; intervals and a suitable comparison are needed.

One-way ANOVA: comparing group means

Do the group means differ overall, and which comparisons answer the research question?

What this method does

One-way ANOVA compares group means for a continuous outcome across the levels of one categorical factor. The overall F test asks whether the means can all be treated as equal.

Required data and design

One continuous outcome, one categorical grouping factor and independent observations.

Outputs to read

Group n, M and SD; F, both df, p, effect size and relevant follow-up comparisons.

What to report

1. Identify the grouping factor and outcome; report n, M and SD for each group.

2. Report F with its between- and within-group degrees of freedom, p and a named effect size such as η^2 or ω^2.

3. If an overall difference is followed up, report the relevant planned or post hoc contrasts, their estimates and intervals, and any multiplicity adjustment.

4. The overall F test does not say that every group differs from every other group. Use the contrasts to state where the differences are.

Worked teaching calculation

If SS_between = 100, df_between = 2, SS_within = 120 and df_within = 12: F = (100/2)/(120/12) = 5; η^2 = 100/220 = .455.

Teaching values for following the calculation; these are not findings from the cited study.

Symbols in this calculation

SS is a sum of squares, df is degrees of freedom and MS = SS/df. F compares the specified mean squares; η^2 is the proportion of total sample variation attributed to the factor in this example.

Interpret the result

The omnibus result concerns at least one contrast; it does not identify the differing groups.

Check the claim and reporting

Specify contrasts and multiplicity; check dependence and variance assumptions.

Research example: how the result is reported

Verbatim excerpt · Original English

> For Developing Argument there was a statistically significant difference between treatment group students and the control group (F = 4.62, df = 1, 27; p = .04), with treatment students making more Developing Argument revisions than the control group (see Table 3). The effect size (Cohen's d) = 0.75.

Early & Saidy (2014). A study of a multiple component feedback approach to substantive revision for secondary ELL and mult
Results · PDF page 12 (checked copy; counted from its first page)

Reading the report

Read the outcome and contrast first, then F, its degrees of freedom and p. The sentence also supplies the direction and Cohen's d; a significant omnibus test alone would not identify every pairwise difference.

Chi-square tests and contingency tables

Are two categorical variables associated, or do category frequencies fit an expected distribution?

What this method does

Compares observed category counts with expected counts under a stated model, including independence in a contingency table.

Required data and design

Counts of independent cases classified by categories.

Outputs to read

Observed/expected counts, χ^2, df, p and association magnitude.

What to report

1. Identify both categorical variables, category coding, independent cases and the contingency-table counts.
2. Report counts, row or column percentages with denominators, and expected-count checks.
3. Report the named test, χ^2, df, N, p and an appropriate association measure such as Cramér's V.
4. Explain the pattern with percentages; identify an exact test if sparse counts require one.

Worked teaching calculation

For [[18,12],[9,21]], expected counts are [[13.5,16.5],[13.5,16.5]]; $\Sigma(O-E)^2/E = 5.45$; df = 1; $\varphi = \sqrt{(5.45/60)} = .302$.

Teaching values for following the calculation; these are not findings from the cited study.

Interpret the result

The category proportions differ in this sample; inspect the cells to explain the pattern.

Check the claim and reporting

Use an appropriate exact procedure for sparse tables; repeated categories require another model.

Research example: how the result is reported

Verbatim excerpt · Original English

> A significant difference was found between the low and high groups, $\chi 2$ (2, N = 120) = 8.75, p < .05, with Cramer's V = 0.27, indicating a medium-to-large effect size (Cohen, 1988, pp. 79-80).

Frost et al. (2021). Relationships between Comprehension, Strategic Behaviours and Content-Related Aspects of Test Perfor
Results · PDF page 10 (checked copy; counted from its first page)

Reading the report

The report combines chi-square, degrees of freedom, N, p and Cramér's V. Inspect the contingency table to see the pattern; the source's effect-size adjective depends on context and should not replace the numerical estimate.

Mann-Whitney, Wilcoxon and Kruskal-Wallis tests

Do independent or paired observations differ under a rank-based hypothesis?

What this method does

Uses ranks to test specified distributional or location hypotheses, with the test chosen for the grouping and pairing.

Required data and design

Ordinal/numerical data with independent or paired structure appropriate to the chosen test.

Outputs to read

Rank statistics, uncertainty and the precise distributional hypothesis.

What to report

1. Choose the test for independent groups or matched observations, and state its hypothesis.
2. Report group sizes, medians and IQRs or other appropriate distribution summaries.
3. Report the test statistic, inferential procedure, p, treatment of ties and a named effect estimate where suitable.
4. Explain what the rank test establishes; a median-only interpretation needs additional distribution assumptions.

Worked teaching calculation

A = 1,2 and B = 3,4: rank sum $R_A = 3$; $U_A = 3-2(3)/2 = 0$; $U_B = 4-0 = 4$.

Teaching values for following the calculation; these are not findings from the cited study.

Interpret the result

The ranks indicate that all A observations are below all B observations in this toy sample.

Check the claim and reporting

Rank tests are not assumption-free and do not generally test a difference in medians.

Research example: how the result is reported

Verbatim excerpt · Original English

> To this end, Mann-Whitney U Test ranks were compared and the results revealed that the integrated assessment group with a mean rank of 18.83 had a better performance on the writing section of PET compared to the non-integrated assessment group with a mean rank of 9.72.

Sadeghi & Rahmati (2017). Integrating assessment as, for, and of learning in a large-scale exam preparation course
Results · PDF page 8 (checked copy; counted from its first page)

Source table extract · Selected values, reformatted

Table 8 · Mann-Whitney U Test statistics (selected entries)

Output	Reported value
Mann-Whitney U	21.500
Z	−2.926
Asymp. Sig. (2-tailed)	.003
Exact Sig. [2*(1-tailed Sig.)]	.003
Mean rank: non-integrated assessment	9.72
Mean rank: integrated assessment	18.83

Selected entries from the source table, rearranged into two columns. Mean ranks are not means on the original writing-score scale.

Sadeghi & Rahmati (2017) · PDF page 8

Reading the report

Read the result in two parts: U = 21.500 and the reported p = .003 indicate evidence of a group difference; the mean ranks identify its direction. The integrated group has the higher mean rank. This comparison does not itself establish that the intervention caused the difference.

Repeated-measures and mixed ANOVA

Do mean outcomes change across occasions, and does change differ by group?

What this method does

Compares means across repeated occasions and, in a mixed ANOVA, examines group-by-occasion interactions.

Required data and design

Repeated scores, occasion identifiers, and optional between-person groups.

Outputs to read

Time, group and time-by-group contrasts; corrected df when required.

What to report

1. Identify the repeated factor, occasions, participants and any between-person groups.

2. Report the same-person sample sizes and occasion means and SDs; show individual trajectories where useful.

3. Report F, both df, p and a named effect size; state sphericity checks and corrections where applicable.

4. Report relevant time contrasts or group-by-time interactions with estimates, intervals and multiplicity adjustments.

Worked teaching calculation

Group A changes 50→60 and B changes 55→58: difference in changes = (60−50)−(58−55) = 7. This contrast still needs its model-based SE.

Teaching values for following the calculation; these are not findings from the cited study.

Interpret the result

The interaction asks whether change differs between groups.

Check the claim and reporting

Significance in one group and not the other is not a test of the interaction.

Research example: how the result is reported

Verbatim excerpt · Original English

> A statistically significant effect of time on the scores after corpus use was found (F(2, 20) = 5.807, p = .017).

Wu (2021). Discovering collocations via data-driven learning in L2 writing
Results · PDF page 8 (checked copy; counted from its first page)

Reading the report

This is an omnibus time effect: it indicates evidence of change somewhere across occasions, rather than showing which occasions differ. Read the pairwise contrasts, effect size and repeated-measures assumptions as well.

ANCOVA

How do group means compare after adjusting for specified covariates?

What this method does

Uses a regression model to compare group means conditional on specified covariates.

Required data and design

Continuous outcome, groups and defensible covariates such as baseline score.

Outputs to read

Adjusted mean difference, coefficient/contrast, interval and covariate specification.

What to report

1. Name the outcome, grouping factor, covariates and why those covariates were included.

2. Report raw group summaries and covariate overlap; examine whether a common-slope comparison is appropriate.

3. Report adjusted means at specified covariate values or a defined averaging distribution, adjusted contrasts and confidence intervals.

4. Report the model test, df, p, effect size and diagnostics. Explain that adjustment alone does not establish causation.

Worked teaching calculation

In $\hat{y} = 10 + 3G + 0.5X$, at baseline $X = 60$: $G = 0$ gives 40; $G = 1$ gives 43; adjusted difference = 3.

Teaching values for following the calculation; these are not findings from the cited study.

Interpret the result

The contrast compares groups at the same specified covariate value.

Check the claim and reporting

Adjustment does not guarantee equivalent groups or identify a causal effect.

Research example: how the result is reported

Verbatim excerpt · Original English

> The LSD post hoc comparisons made on the adjusted mean scores found the perceived automated feedback group (adjusted = 78.953) significantly outperformed the perceived teacher feedback group (adjusted = 73.551) on essay two.

Reynolds et al. (2021). Investigating the Effects of Perceived Feedback Source on Second Language Writing Performance: A Quasi-Experimental Study
Results · PDF page 6 (checked copy; counted from its first page)

Reading the report

The reported means are adjusted for a covariate rather than raw group averages. This is a follow-up contrast; inspect the preceding adjusted model, covariate specification, uncertainty and multiplicity handling.

Linear, multiple and hierarchical regression

How does the expected value of a continuous outcome vary with one or more predictors?

What this method does

Models the conditional mean of a continuous outcome using one or more predictors. Hierarchical entry compares prespecified blocks.

Required data and design

Continuous outcome, predictors and an appropriate error/dependence structure.

Outputs to read

b, SE, intervals, R^2; ΔR^2 for prespecified predictor blocks.

What to report

1. Identify the continuous outcome, predictor units, coding and model specification.

2. Report n, unstandardised b, SE or confidence intervals, test statistics and p for the relevant terms.

3. Report R^2 or adjusted R^2, residual checks and the role of any predictor blocks; distinguish block entry from multilevel modelling.

4. Translate coefficients into outcome units while holding included predictors constant. Distinguish association, prediction and causal claims.

Worked teaching calculation

With $\hat{y} = 20 + 2X + 0.5Z$, $X = 3$ and $Z = 60$ give $\hat{y} = 56$. If $b_X = 2$ and $SE = .5$, an illustrative large-sample 95% Wald interval is $2 \pm 1.96(.5) = [1.02, 2.98]$.

Teaching values for following the calculation; these are not findings from the cited study.

Interpret the result

One extra unit of X is associated with 2 outcome units at the same Z under this model.

Check the claim and reporting

A conditional association and in-sample R^2 do not establish causality or new-sample prediction.

Research example: how the result is reported

Verbatim excerpt · Original English

> The results show that PIA (β = 0.24, p < 0.01), grit (β = 0.23, p < 0.01), self-confidence (β = 0.22, p < 0.01), and motivation (β = 0.15, p < 0.05) were identified as the significant predictors of students' L2 WTC.

Lee & Drajati (2019). Affective variables and informal digital learning of English: Keys to willingness to communicate in a second language
Results · PDF page 6 (checked copy; counted from its first page)

Reading the report

The standardised beta coefficients describe conditional associations within the fitted model. Read their signs, magnitudes and p values, then check intervals and model fit; predictor does not automatically mean cause.

Binary logistic regression

How is the probability of a binary outcome related to predictors?

What this method does

Models the log odds of a binary outcome and can express results as odds ratios or predicted probabilities.

Required data and design

Binary outcome with explicit event coding and predictors.

Outputs to read

Log-odds coefficient, odds ratio, interval, predicted probabilities and calibration.

What to report

1. State the event coded 1, reference categories, sample size and event counts.

2. Report model specification, b or log odds, odds ratios and their confidence intervals, plus relevant p-values.

3. Give predicted probabilities at meaningful predictor values; an odds ratio is not a probability ratio.

4. Report suitable diagnostics and fit or validation evidence, and explain how missing data and sparse events were handled.

Worked teaching calculation

If logit(p) = −1 + .7X: OR = exp(.7) = 2.01. At X = 1, p = 1/(1+exp(.3)) = .426.

Teaching values for following the calculation; these are not findings from the cited study.

Symbols in this calculation

logit(p) = ln[p/(1−p)]; odds = p/(1−p). Exponentiating a log-odds coefficient gives an odds ratio for a one-unit predictor change.

Interpret the result

A one-unit increase in X multiplies the odds by about 2, not the probability.

Check the claim and reporting

Distinguish discrimination from calibration and odds from risks.

Research example: how the result is reported

Verbatim excerpt · Original English

> The results show that TOEFL iBT Speaking scores significantly predicted whether a test taker passed the EPT OC (z = 7.93, p < .001).

Ockey et al. (2021). Establishing Appropriate Cut Scores of a Standardized Test for a Local Placement Context
Results · PDF page 16 (checked copy; counted from its first page)

Reading the report

The outcome is binary: passing the speaking criterion. z and p provide evidence about a coefficient, but an odds ratio or predicted probabilities with intervals are needed to communicate the size of the association.

Internal consistency and rater agreement

How consistent or reproducible are scores under the intended measurement conditions?

What this method does

Evaluates score consistency or agreement with a coefficient matched to the measurement design.

Required data and design

Item-level scores or repeated ratings, depending on the source of inconsistency.

Outputs to read

α/ω, ICC, κ or another justified coefficient with uncertainty and design details.

What to report

1. Describe the scale, scoring, number of items, reverse coding, sample and measurement conditions.

2. Name the coefficient and its version: alpha/omega for suitable internal-consistency questions; an appropriate ICC or kappa for agreement designs.

3. Report the estimate and uncertainty, with the assumptions and missing-item rules used.

4. Discuss what the coefficient supports. High consistency alone establishes neither validity nor a single dimension.

Worked teaching calculation

For k=4 items, sum of item variances=4 and total-score variance=10: $\alpha=(4/3)(1-4/10)=.80$.

Teaching values for following the calculation; these are not findings from the cited study.

Interpret the result

This α describes score consistency under its assumptions in this sample.

Check the claim and reporting

Internal consistency, absolute agreement and rank consistency answer different questions.

Research example: how the result is reported

Verbatim excerpt · Original English

> The Cronbach’s alpha of the whole test (i.e., 30 items) was .79, which was sufficiently high for internal consistency reliability.

Hirai et al. (2022). Development and validation of an English test measuring EFL learners’ critical thinking skills
Results · PDF page 10 (checked copy; counted from its first page)

Reading the report

Alpha is reported for this 30-item test. Internal consistency is not proof of unidimensionality, validity or reliability in every population; examine item structure and the relevant sample.

Mediation and indirect effects

Is an association between X and Y represented by an indirect pathway through M?

What this method does

Represents an indirect association from X to Y through an intervening variable M; causal interpretation needs additional design assumptions.

Required data and design

X, M and Y with a theoretically justified temporal/design structure.

Outputs to read

a, b, direct c′, indirect ab, total effect and interval for ab.

What to report

1. Define X, M and Y, their timing, the path model and the intended associational or causal interpretation.

2. Report a, b, c′ and the indirect estimate ab with clear units and covariate adjustment.

3. Report the interval method for ab; if bootstrapping is used, state the resampling unit, number of resamples and interval type.

4. Report the direct/total estimates where relevant, uncertainty, assumptions and limitations; an indirect effect is not automatic proof of a mechanism.

Worked teaching calculation

For linear models without an X×M interaction: a=.60, b=.40, c′=.10; ab=.24; total=.10+.24=.34. Bootstrap the full product, preserving sampling structure.

Teaching values for following the calculation; these are not findings from the cited study.

Interpret the result

The arithmetic decomposes the specified linear association into direct and indirect components.

Check the claim and reporting

A path product does not establish a causal mechanism, especially with cross-sectional data.

Research example: how the result is reported

Verbatim excerpt · Original English

> The indirect effects of the three formative assessment strategies via growth mindset were not significant: sharing learning progressions (ß = .004, SE = .67, p = .355), providing feedback (ß = .002, SE = .59, p = .615), and instructional adjustments (ß = .01, SE = .71, p = .198).

Yan et al. (2021). Formative assessment, growth mindset, and achievement: examining their relations in the East and the
Results · PDF page 14 (checked copy; counted from its first page)

Reading the report

The authors identify the indirect pathway and report coefficients, SEs and p values. Check the coefficient and SE scales in the full model and look for an indirect-effect interval; a nonsignificant estimate does not establish no mediation.

Moderation and interaction models

Does the association of X with Y differ across values of W?

What this method does

Includes an interaction to examine whether the association between X and Y changes with W.

Required data and design

Outcome, focal predictor, moderator and their product term with appropriate coding.

Outputs to read

Interaction coefficient, conditional slopes, predicted values and intervals.

What to report

1. Define X, Y and W, coding or centring, and the interaction term.

2. Report the interaction coefficient with units, confidence interval and p, alongside the lower-order terms.

3. Report conditional slopes or predicted outcomes at meaningful W values and show an interaction plot.

4. Compare slopes through the interaction; separate significant/non-significant subgroup results do not themselves establish a difference.

Worked teaching calculation

If $\hat{y}=2+.4X+.3W+.2XW$, slope_X=.4+.2W. At W=−1,0,1 the slopes are .2,.4,.6.

Teaching values for following the calculation; these are not findings from the cited study.

Interpret the result

The model predicts a stronger X-Y association at higher W.

Check the claim and reporting

Test the interaction directly; do not infer it from unequal subgroup p-values.

Research example: how the result is reported

Verbatim excerpt · Original English

> Regarding moderation, GenAI literacy weakened the effect of social influence on behavioral intention (β = −0.094, p = 0.039, f2 = 0.013).

Wang & Zhang (2026). What Drives GenAI Adoption in Informal Digital Learning of English (IDLE)?: Structural□Configurational Modelling of Extended UTAUT2 with GenAI Literacy
Results · PDF page 6 (checked copy; counted from its first page)

Reading the report

The coefficient concerns an interaction: the association changes with GenAI literacy. Interpret it using conditional slopes or predictions; f^2 supplies an effect-size perspective alongside p.

Quantitative data analysis: method map

We have practised frequently used methods together. Use this map to explore other methods at your own pace: find your question and data type, open a method, then return here to compare another. The next lesson page is the glossary.

Main categorisation reference: UCLA OARC · Choosing the Correct Statistical Test. Advanced branches extend the map by question and data structure; each choice still requires checks of the model assumptions.

Choose by outcomes and predictors

Describe one sample

What does the sample distribution show?

Outcomes

One or more · Continuous or categorical

Predictors

None required

- Descriptive statistics

One outcome: compare groups

Do group means or rank distributions differ?

Outcomes

One · Continuous or ordinal

Predictors

Categorical groups; optional covariates

- Independent and paired t-tests
- One-way and factorial ANOVA
- ANCOVA
- Mann-Whitney, Wilcoxon and Kruskal-Wallis tests
- Equivalence testing / TOST

One categorical outcome

Are categories associated, or what predicts category probabilities?

Outcomes

One · Binary, ordered or unordered categories

Predictors

Categorical, continuous or mixed

- Chi-square tests and contingency tables
- Binary logistic regression
- Ordinal and multinomial regression

One continuous outcome: association or regression

How does a continuous outcome vary with the predictors?

Outcomes

One · Continuous

Predictors

Continuous, categorical or mixed

- Pearson, Spearman and partial correlation
- Linear, multiple and hierarchical regression
- Generalised additive models (GAMs)
- Robust regression

Several continuous outcomes

Do groups differ across outcomes considered jointly?

Outcomes

Two or more · Several continuous outcomes

Predictors

Categorical groups; optional covariates

- MANOVA and MANCOVA

Many indicators: dimensions and measurement

What structure underlies the indicators?

Outcomes

No single designated outcome · Continuous or categorical indicators

Predictors

Several observed indicators

- Exploratory factor analysis
- Confirmatory factor analysis
- Principal component analysis
- Measurement invariance / multi-group CFA
- Internal consistency and rater agreement
- Item response theory: 2PL and polytomous models

Repeated outcomes across time

How do outcomes change or relate across occasions?

Outcomes

One or more, repeatedly measured · Repeated continuous outcomes or time series

Predictors

Time, groups or prior measurements

- Repeated-measures and mixed ANOVA
- Latent growth curve modelling
- Cross-lagged panel models
- Random-intercept CLPM
- Time-series analysis

Extend for the question and data structure

Indirect effects, interactions and SEM

Through which pathway, or under which conditions, are variables related?

Outcomes

One or a system of outcomes · Observed or latent outcomes

Predictors

Predictors, mediators, moderators or latent constructs

- Mediation and indirect effects
- Moderation and interaction models
- Moderated mediation / conditional process
- Structural equation and path modelling

Event counts and rates

What predicts the expected count or rate?

Outcomes

One · Non-negative counts; exposure if needed

Predictors

Continuous, categorical or mixed

- Count regression (Poisson / negative binomial)

Clusters, latent profiles and latent classes

Which cases show similar multivariate patterns?

Outcomes

No designated outcome for grouping · Continuous for LPA; categorical for LCA

Predictors

Several grouping indicators

- K-means and hierarchical clustering
- Latent profile analysis
- Latent class analysis (LCA)

Nested, crossed or otherwise correlated observations

How should the model represent dependence?

Outcomes

One or more correlated outcomes · Match the response distribution and link

Predictors

Predictors plus person/group/item structure

- Linear / generalised mixed-effects models

- Generalised estimating equations

Time until an event

How long until an event occurs?

Outcomes

One event-time outcome · Time plus event/censoring status

Predictors

Continuous, categorical or mixed

- Survival / time-to-event analysis

Conditional relationships in a system

Which variables remain associated conditional on the others?

Outcomes

No single designated outcome · Joint multivariate observations

Predictors

Variables treated as nodes

- Network analysis: conditional-association models

Prediction for unseen cases

How accurate are predictions on held-out data?

Outcomes

One or more · Continuous or categorical

Predictors

Features with independent validation

- Random forests and boosted trees

- Support vector machines

- Neural networks for prediction

Effect estimates from several studies

What do study estimates show together?

Outcomes

Study effect estimates · Effect estimates and sampling variances

Predictors

Study characteristics or none

- Random-effects meta-analysis
- Meta-regression and moderator analysis
- Three-level meta-analysis / robust variance estimation

Estimation and uncertainty across models

How will parameters and uncertainty be estimated?

Outcomes

Defined by the substantive model · Must respect the model and sampling structure

Predictors

Defined by the substantive model

- Bayesian estimation
- Bootstrap / resampling inference

Statistical glossary

101 terms: start with describing and comparing data, continue with common reporting notation, then consult terms for particular analyses. Definitions are written out within each category.

Click − to collapse a category or + to expand it. Several categories can stay open together.

The third column distinguishes mathematical bounds, no-effect references and conventional benchmarks. Some terms have no numerical range or universal cutoff; interpretation depends on the model, scale, design and research context.

Start with describing and comparing

Describing data

Term / notation	Meaning	Range and numerical interpretation
n: sample size	The number of observations or units included in the stated analysis. Check the unit: 300 ratings from 30 people do not represent 300 independent people.Also written as: n · N	**Positive whole number**n = 80 means 80 stated units. Identify whether these are people, responses or studies; there is no universal minimum sample size.
$\bar{x}$ / M: sample mean	Add the observed values and divide by their number. For 2, 4, 4, 6, the mean is 16/4=4. In descriptive tables M often means mean; in a mediation diagram M instead labels the mediator.Also written as: mean · sample mean · $\bar{x}$ · □□□ · M	**Within the measurement scale**A mean on a 1-5 scale lies between 1 and 5. It can be negative if the scale permits negative values; there is no universal “good” mean.
Median	The middle value after sorting observations, or the mean of the middle two when their number is even. It describes location and can differ from the mean in a skewed distribution.Also written as: median · □□□	**Within the measurement scale**At least 50% of the distribution is at or below the median and at least 50% at or above it. Its possible numerical values follow the score scale.
SD: standard deviation	Describes the spread of observations around their mean, in the original measurement units. SD is the square root of variance. It is different from the standard error of an estimate.Also written as: standard deviation · standard deviations · SD · □□□	**SD ≥ 0**0 means no observed spread. SD = 2 means a spread measured in two original score units; its size is meaningful relative to the scale.
Variance	A measure of spread based on squared deviations. Sample variance usually divides the sum of squared deviations from the sample mean by n−1. Variance uses squared units; SD returns to the original units.Also written as: variance · variances · □□□	**Variance ≥ 0**0 means no variation. Variance = 4 corresponds to SD = 2; variance uses squared units. A negative fitted variance component requires investigation.
IQR / Q1 / Q3	The interquartile range is Q3−Q1: the spread of the middle 50% of observations. Q1 and Q3 are the 25th and 75th percentiles. IQR describes observations, rather than the uncertainty of a mean estimate.Also written as: IQR · quartile · Q1 · Q3 · □□□□	**IQR ≥ 0; Q1 ≤ Q3**Q1 and Q3 mark the 25th and 75th percentiles. IQR = 2 means the middle half of observations spans two scale units.
Range / minimum / maximum	Minimum and maximum are the smallest and largest observed values. Range=maximum−minimum. They describe the observed sample; they are different from the lower and upper limits of a confidence interval.Also written as: range · minimum · maximum · □□	**Range ≥ 0; minimum ≤ maximum**With minimum 2 and maximum 8, range = 6. Minimum and maximum can be negative; they are not confidence limits.

Term / notation	Meaning	Range and numerical interpretation
σ² / σ: variance and SD parameters	σ² often denotes a population or model variance, and σ its standard deviation. Subscripts specify the component, such as residual variation. Check the model's units and component definitions.Also written as: σ² · σ	**σ ≥ 0; σ² ≥ 0**σ = 2 corresponds to σ² = 4. There is no shared upper bound across measurement scales.
μ: expected value	A model mean or expected value. In count regression μ is the expected count; in growth models μ with a subscript may denote a mean intercept or mean slope.Also written as: μ	**Follows the outcome/model domain**A Gaussian mean can be any real value; a count mean is non-negative, and a binary-response mean is a probability between 0 and 1.
Σ: sum	Add the indicated quantities over the specified index. For example, Σx adds all the x values. The index and limits identify which observations or components are included.Also written as: Σ	**No fixed range**Σ is an operation, not one statistic with a cutoff. A sum of signed values can be negative; a sum of squares cannot.
Standardised	Expressed on a rescaled metric, often using standard deviations. A standardised variable z=(x−mean)/SD has mean 0 and SD 1 in the sample used for the transformation. State which variables or coefficients were standardised.Also written as: standardised · standardized · □□□	**Depends on the quantity standardised**Standardising does not generally confine values to −1 to +1. For z scores, 0 is the mean and 1 is one SD above it.
z: standardised score	Here z denotes a standardised variable: (value−mean)/SD. A value of 1 is one SD above the mean; −.5 is half an SD below it. This is different from a coefficient's z-test statistic.Also written as: z□ · z□	**Any real value: −∞ to +∞**0 = sample mean; +1/−1 = one SD above/below. For an approximately normal distribution, about 95% fall within ±1.96; these are not absolute score limits.NIST: normal critical values

Comparing groups

Term / notation	Meaning	Range and numerical interpretation
Mean difference	The difference between two specified means. Define its direction: mean(A)−mean(B)=3 means A averages three outcome units higher. For paired data, analyse within-pair differences and their uncertainty.Also written as: mean difference · □□□	**Can be negative, zero or positive**For A−B, +3 means A is three units higher; −3 means lower. 0 is the no-difference reference; practical importance depends on the scale and interval.
t statistic	For a coefficient or mean difference, t=(estimate−null value)/SE. Compare it with the appropriate t distribution and degrees of freedom. In time-indexed equations, t can instead label an occasion.Also written as: t	**Any real value**0 is the null-centred reference. Sign follows the contrast; larger |t| gives a smaller two-sided p at fixed df. Critical values depend on df and the chosen α, not one universal t cutoff.
F statistic	A ratio of model-defined mean squares or a joint-test statistic. In a simple ANOVA, F=MS_between/MS_within. Read both numerator and denominator df; a significant omnibus test does not identify the differing groups.Also written as: F	**F ≥ 0**In ordinary ANOVA, F near 1 means comparable effect and error mean squares. The upper-tail cutoff depends on both df and α; F is not a percentage or an effect size.
df: degrees of freedom	An index used in the statistic's reference distribution, reflecting information remaining after constraints or fitted parameters. For a paired t-test on n differences, df=n−1. It is not generally the sample size.Also written as: degrees of freedom · df · □□□	**Usually ≥ 0; can be non-integer**Paired t: df = n−1. Welch approximations may use fractional df. SEM df = 0 denotes a saturated/just-identified specification whose global fit cannot test restrictions.

Term / notation	Meaning	Range and numerical interpretation
Effect size	A defined measure of the magnitude of a difference or association. Examples include mean differences, standardised differences and correlations. Interpret the units, uncertainty and substantive context alongside p-values.Also written as: effect size · effect sizes · □□□	**Depends on the effect-size definition**The no-effect reference may be 0 (mean difference, r, d) or 1 (OR, IRR, HR). There is no numerical cutoff shared by all effect sizes.
d: standardised mean difference	A mean difference divided by a specified SD. d=.50 means half of that SD, rather than a 50% improvement. Paired and independent-group versions use different standardisers, which must be stated.Also written as: Cohen d · Cohen's d · Cohen's d · standardised mean difference	**Signed; no universal ±1 bound**0 = no standardised mean difference. Conventional |d| anchors .20/.50/.80 are called small/medium/large; they are rough context-dependent benchmarks, not significance cutoffs.Cohen: conventional effect sizes
g: Hedges' g	A standardised mean-difference estimator with a small-sample bias correction. State the effect-size definition, comparison direction, uncertainty and treatment of dependent data.Also written as: Hedges g · Hedges' g · Hedges' g	**Signed; no universal ±1 bound**0 is the no-difference reference. The .20/.50/.80 standardised-difference anchors are sometimes applied, but g includes a bias correction and should be interpreted with its interval and field context.Cohen: conventional effect sizes
χ^2: chi-square statistic	A statistic evaluated against a specified chi-square reference distribution. In a contingency table it summarises discrepancies between observed and expected counts; in SEM it assesses model discrepancy. Its df and model context matter.Also written as: χ^2 · chi-square statistic · □□□□□	**$\chi^2 \geq 0$ for the usual statistic**0 means no observed discrepancy on that statistic. Larger values indicate more discrepancy; compare with the df-specific reference distribution, not a universal number such as 3.84.
SS: sum of squares	A sum of squared deviations allocated to a model component. Dividing SS by its df gives a mean square (MS). The subscripts identify the component, such as between or within groups.Also written as: SS	**SS ≥ 0 for a sum of squares**0 means all contributing deviations are zero. Raw SS grows with scale and sample size, so it has no common "large" cutoff.
MS: mean square	A sum of squares divided by its associated degrees of freedom: MS=SS/df. ANOVA F ratios compare particular mean squares.Also written as: MS	**MS ≥ 0 when df > 0**MS = SS/df. It is in squared outcome units; compare the appropriate mean squares rather than interpreting one MS against a fixed threshold.
η^2: eta squared	An ANOVA effect-size measure: SS_effect/SS_total. It expresses the proportion of sample total variation attributed to that effect. Partial η^2 uses a different denominator.Also written as: η^2	**$0 \leq \eta^2 \leq 1$ for the usual SS ratio**η^2 = .06 means 6% of the sample total variation is attributed to the specified effect. Rough .01/.06/.14 benchmarks are sometimes labelled small/medium/large; design and context matter.
ηp^2: partial eta squared	SS_effect/(SS_effect+SS_error) for the specified effect and error term. It uses a different denominator from η^2 and is not generally the proportion of all outcome variance explained.Also written as: partial eta squared · partial η^2 · ηp^2 · □η^2	**$0 \leq \eta p^2 \leq 1$ for the usual SS ratio**ηp^2 = .06 means 6% of effect-plus-associated-error SS, not 6% of all outcome variation. Values depend on the design; avoid transferring size labels mechanically across designs.

Frequently used in Results sections

Common reporting: p, SE and confidence intervals

Term / notation	Meaning	Range and numerical interpretation
p-value	Under the null hypothesis and specified model, the probability of a test statistic at least as extreme as the observed one. p=.02 means 2% under those assumptions; it is not a 2% probability that the null is true, and does not measure effect size. The notation p < .001 means this tail probability is below 0.1%, under the null and specified model; it is not the probability that the null hypothesis is true.Also written as: p-value · p-values · p value · p · p□	**0 ≤ p ≤ 1**p < .05 / .01 / .001 means a null-model tail probability below 5% / 1% / 0.1%. These are conventional reporting thresholds; compare with the prespecified α. Neither small p nor p ≥ .05 establishes effect size or proves a hypothesis.ASA: interpreting p-values
SE: standard error	The estimated sampling variability of an estimate. SE describes precision of the estimate, whereas SD describes spread among observations. A simple large-sample 95% Wald CI is estimate ± 1.96×SE. SE = 0.50 means the estimated standard error is 0.50 in the estimate's units; for a regression coefficient those units are outcome units per predictor unit.Also written as: standard error · standard errors · SE · □□□	**SE ≥ 0; no fixed upper bound**Smaller SE means greater precision for the same parameter and units. SE is not judged against .05; that threshold concerns p. SE = 0 may indicate a degenerate or numerical situation.
CI: confidence interval	A lower and upper limit expressing uncertainty about a parameter. Under its assumptions, a 95% CI procedure covers the true parameter in 95% of repeated samples. [1.02, 2.98] lists lower=1.02 and upper=2.98; it is not a range containing 95% of individuals. With an estimate of 2.00 and SE of 0.50, a large-sample normal-reference Wald calculation gives 2.00 ± 1.96 × 0.50 = [1.02, 2.98]. A t-based interval instead uses a multiplier determined by the degrees of freedom.Also written as: confidence interval · confidence intervals · CI · □□□□	**Limits follow the parameter's scale**Read the confidence level (often 95%) and both limits. For a matched two-sided 5% test, a 95% CI excluding the null corresponds to rejection: null = 0 for differences/slopes, 1 for ratios. This correspondence requires the same inferential method.
[lower, upper]: interval notation	In 95% CI [1.02, 2.98], 1.02 is the lower limit and 2.98 is the upper limit, in the units of the parameter. These are interval endpoints, rather than two separate estimates or the range of 95% of individual scores. Read the label: square brackets can also show equivalence bounds or other intervals.Also written as: square brackets · [lower, upper] · □□□	**Lower limit ≤ upper limit**[1.02, 2.98] means lower 1.02 and upper 2.98. For a coefficient the interval excludes 0; for a ratio it includes 1 only if 1 lies between its limits. Always identify the labelled parameter and interval type.
LL / lower limit	The lower endpoint of the labelled interval. In CI [1.02, 2.98], LL=1.02. The lower limit is not the smallest observed individual score.Also written as: LL · lower limit · □□	**LL ≤ UL; sign follows the parameter**A negative lower limit is possible for a signed coefficient. A ratio's admissible limits are non-negative; LL itself has no universal pass/fail value.
UL / upper limit	The upper endpoint of the labelled interval. In CI [1.02, 2.98], UL=2.98. Read it together with the lower limit, estimate and stated confidence level.Also written as: UL · upper limit · □□	**UL ≥ LL; sign follows the parameter**Read together with LL. A narrow interval indicates greater precision on the same scale; a large positive UL alone does not establish an effect.

Term / notation	Meaning	Range and numerical interpretation
z statistic	A common Wald z statistic is (estimate − null value)/SE. Its magnitude expresses distance from the null in standard-error units. A reference standard normal distribution is used when the approximation is justified. z is not an effect size. Against a null coefficient of 0, an estimate of 2.00 with SE = 0.50 gives z = (2.00−0)/0.50 = 4.00: four standard errors from zero. In a neural network z can instead mean the weighted input plus bias before activation; a standardised observation is another use of z.Also written as: z	**Any real value**For a justified standard-normal two-sided test: |z| ≈ 1.960 / 2.576 / 3.291 corresponds to p ≈ .05 / .01 / .001. These are reference-specific critical values, not effect-size labels.NIST: normal critical values
p = .000 in an output	Usually a display rounded to three decimal places, not a probability known to be exactly zero. Use a justified reporting convention such as p<.001 when the underlying value supports it.Also written as: p=.000 · p = .000 · rounding · □□□□	**Rounded display, not a special p range**p = .000 usually means the display has rounded a small positive value. Report p < .001 when supported; do not interpret it as zero probability.ASA: interpreting p-values
H□: null hypothesis	The reference claim tested by the specified procedure, for example a mean difference of 0 or an odds ratio of 1. The null value depends on the parameter and the test.Also written as: H0 · H□ · null hypothesis · □□□	**A claim, not a bounded statistic**Common null values: difference/slope/r = 0; ratio = 1. Define the actual parameter and null before looking at p.
H□ / H_a: alternative hypothesis	The specified alternative to H□: for example a difference other than 0, or a difference greater than 0. A significant result does not assign a probability that this alternative is true.Also written as: H1 · H□ · H_a · alternative hypothesis · □□□□	**A claim, not a bounded statistic**A two-sided alternative may be b ≠ 0; a directional alternative b > 0. A small p does not provide P(H□ is true).
α: significance level	A prespecified long-run Type I error threshold for the test, often .05. Compare p with the chosen threshold, while still interpreting magnitude, uncertainty, multiplicity and design. This α differs from reliability alpha.Also written as: significance level · □□□□	**0 < α < 1 in ordinary testing**Common choices .05, .01 and .001 target long-run Type I error rates of 5%, 1% and 0.1% under the test assumptions. Choose α in advance, accounting for multiple testing where needed.ASA: interpreting p-values
Critical value / ±	A reference-distribution cutoff associated with the chosen tail probability. In estimate ±1.96×SE, ± means compute both minus and plus endpoints. The value 1.96 is a normal-reference multiplier; a t-based interval uses a df-specific multiplier.Also written as: critical value · ± · □□□	**Depends on distribution, tails, df and α**Normal reference: 95% two-sided uses ±1.96; a 5% upper one-sided test uses +1.645. A t test uses a df-specific value. ± instructs calculating both minus and plus limits.NIST: normal critical values

Correlation and regression

Term / notation	Meaning	Range and numerical interpretation
r: correlation	Pearson r expresses linear association on a scale from −1 to +1. Sign gives direction; magnitude indicates strength of that linear pattern. A correlation does not establish causality.Also written as: r	**−1 ≤ r ≤ +1**0 = no linear association; ±1 = perfect linear association. Conventional |r| anchors .10/.30/.50 indicate small/medium/large in some settings; they are not universal field standards or p-value thresholds.Cohen: conventional effect sizes

Term / notation	Meaning	Range and numerical interpretation
R^2: coefficient of determination	In ordinary linear regression, the proportion of sample outcome variation accounted for relative to an intercept-only model. R^2=.40 means 40% on that definition. Other models use different pseudo-R^2 measures.Also written as: R^2 · R-squared	**Usually 0-1 for in-sample OLS with an intercept**R^2 = .40 means 40% of sample outcome variation accounted for on that definition. Predictive/out-of-sample or no-intercept R^2 can be negative. No universal "good R^2" applies across tasks.scikit-learn: predictive R^2
Coefficient	A model parameter describing a specified relationship. Read its sign, size, units and uncertainty. A coefficient is not automatically a causal effect.Also written as: coefficient · coefficients · □□	**Usually signed; range depends on the model**Read the defined scale. 0 often means no conditional association on the coefficient scale; size is not judged by p cutoffs or one standard numerical benchmark.
b: unstandardised coefficient	In linear regression, b is the expected outcome-unit difference per one-unit predictor difference, holding the other included predictors constant. For example, b=2 means two outcome units per predictor unit. Its meaning changes with the model's response/link scale. Other uses in this lesson: IRT b=item difficulty; mediation b=M→Y path conditional on X; SVM b=intercept/bias; GAM b_k=basis-function coefficient. These meanings should not be interchanged.Also written as: b	**Can be negative, zero or positive**No fixed upper bound or general cutoff. b = 2 means two outcome units per predictor unit in a linear model; b = −2 reverses direction. IRT difficulty and other uses require their own scale.UCLA: regression coefficients
β	In many regression outputs, β denotes a standardised coefficient: the outcome SD difference per predictor SD difference, conditional on the other predictors. In equations β may instead denote a population coefficient. Check the stated notation. In power calculations β is a Type II error probability. Beta(a,b), written as a distribution name, denotes the beta probability distribution, as in the Bayesian example.Also written as: β · beta	**Signed; not generally bounded by ±1**A standardised β = .30 means .30 outcome SD per predictor SD, conditional on other predictors. In multiple regression |β| can exceed 1; do not apply correlation cutoffs automatically. Type II error β instead lies between 0 and 1.UCLA: regression coefficients
Intercept	The model's expected outcome, or linear predictor, when numeric predictors equal zero and categorical predictors are at their reference levels. Its interpretation depends on coding and centring.Also written as: intercept · □□	**Can be negative, zero or positive on the model scale**0 is the model prediction at the coded zero/reference conditions, not a goodness threshold. An intercept on a logit scale needs transformation to obtain probability.
Slope	The modelled change in the outcome or link scale per unit of a predictor. With an interaction, the slope can depend on another variable; with growth models it often represents change per time unit.Also written as: slope · slopes · □□	**Usually any real value**Positive = increasing modelled outcome/link value; negative = decreasing. A slope of 0 is the no-change/no-association reference, conditional on the specified model.
X: predictor variable	X names a predictor in these equations. A one-unit X difference is interpreted on X's stated scale; coding matters when X represents a group. X is a variable label rather than a test statistic.Also written as: X	**Variable label: no inherent numerical range**The range belongs to X's measurement or coding: for example 0/1 groups, a 1-5 score or unrestricted values. The letter itself has no cutoff.
Y: outcome variable	Y names the outcome or response being modelled. In a repeated-measures equation, a subscript identifies the occasion or person. Y is observed; $\hat{y}$ denotes a model prediction.Also written as: Y	**Variable label: no inherent numerical range**The range belongs to Y's measurement or coding: for example 0/1 groups, a 1-5 score or unrestricted values. The letter itself has no cutoff.

Term / notation	Meaning	Range and numerical interpretation
Z: context-dependent symbol	Z can name an additional predictor, a standardised score or a normal-reference statistic. Read the equation or output label before assigning it a meaning.Also written as: Z	**Variable label: no inherent numerical range**The range belongs to Z's measurement or coding: for example 0/1 groups, a 1-5 score or unrestricted values. The letter itself has no cutoff.
IV: independent/predictor variable	A variable used to explain or predict an outcome in the model. Calling it independent does not mean it is statistically independent of other predictors or experimentally manipulated.Also written as: IV · independent variable · predictor variable · □□□ · □□□□	**Variable label: no inherent numerical range**The range belongs to the predictor's measurement or coding: for example 0/1 groups, a 1-5 score or unrestricted values. The letter itself has no cutoff.
DV: dependent/outcome variable	The response being described, compared or predicted. Its type-continuous, binary, ordinal, count or time-to-event-is central to choosing a model. "Dependent variable" is a role label, not proof of causal dependence.Also written as: DV · dependent variable · outcome variable · □□□ · □□□□	**Variable label: no inherent numerical range**The range belongs to the outcome's measurement or coding: for example 0/1 groups, a 1-5 score or unrestricted values. The letter itself has no cutoff.
ŷ: predicted outcome	The model's fitted or predicted value of Y at the stated predictor values. The hat distinguishes a prediction from an observed outcome; an individual observation can differ from this value.Also written as: ŷ	**Follows the fitted model's output scale**A linear prediction may be any real value; a predicted probability must be 0-1, and a predicted count non-negative. Out-of-range linear predictions can flag a mismatch to the outcome.
Residual	For an observed continuous outcome, the observed value minus its model-fitted value. Residual patterns help diagnose model problems; a residual is different from a regression coefficient.Also written as: residual · residuals · □□	**Can be negative, zero or positive**Observed − predicted: +2 means underprediction by two units; −2 means overprediction. 0 means an exact fitted value for that observation; raw residuals have no universal ±2 cutoff.
Reference category / group	The category against which coded coefficients are compared. If group A is the reference and b for B is 3, the modelled B−A contrast is three units on the stated response/link scale.Also written as: reference category · reference group · □□□	**A category/coding choice, not a statistic**A dummy-coded reference is often coded 0, while the comparison is 1. These numbers are labels, not evidence that one group is better.
Covariate	An additional variable included in a model to specify a conditional comparison or relationship. Adjustment does not automatically remove confounding or establish causality.Also written as: covariate · covariates · □□□	**Range follows the covariate's measurement**Age may be non-negative; a centred score may be negative. No numerical cutoff determines whether a variable should be adjusted for; the research design and target comparison do.
Centring	Subtract a chosen reference, often the sample mean, from a predictor. This changes what zero represents and can make intercepts and interaction main terms easier to interpret. It does not automatically solve all collinearity problems.Also written as: centring · centering · □□□	**Centred values can have either sign**Mean-centred 0 means the sample mean, not absence of the trait. +2/−2 means two original units above/below the chosen reference.
VIF: variance inflation factor	A diagnostic of predictor collinearity in a specified regression design. $VIF_j=1/(1-R^2_j)$, where R^2_j comes from regressing predictor j on the others. Interpret it alongside model purpose and design; a single cutoff is not a complete diagnosis.Also written as: VIF · collinearity · □□□	**VIF ≥ 1 for the usual intercept-based calculation**1 means no linear prediction by the other predictors. 5 or 10 are common warning heuristics, not automatic deletion rules. Interpret precision, sample size and the specified design.O'Brien: VIF cautions

Data types and observation structure

Term / notation	Meaning	Range and numerical interpretation
Continuous outcome	An outcome treated as numerical on a scale where differences in magnitude are meaningful. Choosing a linear model still requires examining its form, residual behaviour and dependence.Also written as: continuous outcome · continuous outcomes · □□□□	**Type or structure: no universal numeric range**Values follow the measurement scale; 0 is meaningful only if defined by that scale.
Categorical outcome	An outcome taking distinct categories. Binary has two categories; ordinal has a meaningful order; nominal categories have no assumed order. The response type affects the appropriate model.Also written as: categorical outcome · categorical outcomes · □□□□	**Type or structure: no universal numeric range**Binary labels are often 0/1. Ordered category labels such as 1-5 do not automatically have equal spacing.
Independent observations	Under the specified model, information about one observation does not change the distribution of another. Different people can still share a cluster. Independence is a property justified by the sampling/design and model, not by having different names.Also written as: independent observations · independent groups · □□□□ · □□□□	**Type or structure: no universal numeric range**Independence is a design/model property, not established by a particular correlation p-value.
Paired / matched observations	Two linked measurements, such as the same person before and after, or deliberately matched units. A paired t-test evaluates the mean of the within-pair differences and uses their SD and SE.Also written as: paired · matched pairs · □□	**Type or structure: no universal numeric range**Each pair links two measurements. Pair identifiers are labels; analyse within-pair differences using their scale.
Repeated observations	Multiple measurements come from the same unit, such as one student assessed three times. Preserve that linkage in the analysis rather than treating each row as an unrelated person.Also written as: repeated observations · repeated measurements · □□□□ · □□□□	**Type or structure: no universal numeric range**The number and spacing of occasions must be stated. Three scores from one person are not three independent people.
Nested observations	Each lower-level unit belongs to one higher-level unit in the design: for example, students within classes. Shared class conditions can induce dependence. A model may need to distinguish within-class and between-class variation.Also written as: nested observations · nested · □□□□ · □□	**Type or structure: no universal numeric range**Cluster labels have no numerical ordering; represent the lower/higher-level structure explicitly.
Crossed observations	Two grouping factors overlap instead of one fitting entirely inside the other. For example, several raters score several essays: ratings can share a rater, an essay, or both.Also written as: crossed observations · crossed · □□□□ · □□	**Type or structure: no universal numeric range**Rater and item identifiers are labels, not magnitudes. An observation may share either grouping factor.
k: number or index	k often denotes a number of groups, items, clusters or parameters, or indexes one component. Its definition is local to the model. It is not a universal test statistic.Also written as: k · K	**Usually a non-negative/positive integer or index**k = 3 may mean three groups, items or parameters; it has no fixed meaning or cutoff across models. State the local definition.

Use when the analysis calls for it

Further test terminology: tails, errors and power

Term / notation	Meaning	Range and numerical interpretation
One-sided test	Tests a prespecified directional alternative. The direction should follow the research question and plan, rather than being chosen after seeing which direction produces a smaller p-value.Also written as: one-sided · one-tailed · □□	**A test choice, not a statistic**A 5% upper-tail normal test uses z > 1.645; a lower-tail test uses z < −1.645. The direction must be specified before examining the result.NIST: normal critical values
Two-sided test	Allows departures from the null in either direction. In a simple symmetric normal-reference test with α=.05, the critical values are approximately −1.96 and +1.96.Also written as: two-sided · two-tailed · □□	**A test choice, not a statistic**At α = .05 a standard-normal two-sided test allocates .025 to each tail; critical values are approximately ±1.96. Other reference distributions differ.NIST: normal critical values
Type I error	Rejecting a specified null hypothesis when it is true. The significance level controls this probability under the procedure's assumptions, rather than the probability that a particular published conclusion is wrong.Also written as: Type I error · □□□□□	**An error event; its probability is 0-1**A prespecified α = .05 targets a 5% long-run false-rejection rate under the null and the procedure's assumptions. It is not a 5% probability that this conclusion is wrong.ASA: interpreting p-values
β: Type II error probability	Failing to reject H□ under a specified alternative. This β depends on effect size, sample size, variability and the test; it is a different meaning of β from a standardised regression coefficient.Also written as: Type II error · □□□□□	**0 ≤ β ≤ 1**β = .20 means a 20% chance of not rejecting H□ under the specified alternative; power is then .80. This is not the regression coefficient β.
Power = 1−β	The probability that a specified test rejects H□ under a specified alternative. Power=.80 means 80% in that scenario. Power is not the probability that a non-significant result proves no effect.Also written as: power · 1−β · □□□	**0 ≤ power ≤ 1**.80 and .90 are common planning targets: 80% or 90% rejection probability under the specified alternative. They require a stated effect, α, design and sample size.Cohen: conventional effect sizes

Probabilities, odds, counts and event timing

Term / notation	Meaning	Range and numerical interpretation
P / p: event or response probability	In probability equations P or p denotes a probability between 0 and 1. In logistic regression p=.426 means a 42.6% modelled event probability; it is not a hypothesis-test p-value. P(A|B) means the probability of A conditional on B.Also written as: P · p · event probability · predicted probability	**0 ≤ P ≤ 1**0 = impossible under the model; 1 = certain; .50 = 50%. A predicted probability of .05 is not a p-value or a significance decision.
OR: odds ratio	A ratio of odds, not probabilities. In a logistic model, exp(b) gives the odds ratio per one-unit predictor difference with other included predictors held constant. OR=1 is the no-association reference; OR=2 does not generally mean twice the probability.Also written as: odds ratio · odds ratios · OR · □□□	**OR > 0 for finite log-odds contrasts**1 = equal odds; 2 = twice the odds; .50 = half the odds. These are not probability ratios. For inference, compare a corresponding interval with 1.
Logit	logit(p)=ln[p/(1−p)], the log of the odds. Logistic regression models this scale, so a coefficient on the logit scale is not directly a probability difference.Also written as: logit	**−∞ to +∞ for 0 < p < 1**logit = 0 corresponds to p = .50; +1 to about .731; −1 to about .269. Values approach infinities as p approaches 1 or 0.

Term / notation	Meaning	Range and numerical interpretation
exp: exponential function	exp(x)=eˣ. Exponentiating a logistic-regression coefficient gives an odds ratio; exponentiating a count-model log-rate coefficient gives a rate ratio.Also written as: exp	**exp(x) > 0 for finite x**exp(0) = 1; exp(.693) ≈ 2. A negative log coefficient produces a ratio below 1, not a negative ratio.
IRR: incidence-rate ratio	For a count model with a log link, exp(b) expresses a multiplicative change in the expected count, or rate when exposure is appropriately incorporated. IRR=1.35 means a 35% higher modelled count/rate, conditional on included predictors.Also written as: IRR · incidence-rate ratio · □□□□	**IRR > 0 for finite log-rate contrasts**1 = equal modelled counts/rates; 1.35 = 35% higher; .70 = 30% lower. For rates, exposure must be included appropriately; interval reference is 1.
HR: hazard ratio	A ratio of event hazards under the specified survival model, conditional on being event-free up to that time. HR=1.5 does not directly mean the cumulative probability of the event is 50% higher.Also written as: HR · hazard ratio · □□□□	**HR > 0 for finite log-hazard contrasts**1 = equal hazards; 1.50 = 50% higher hazard among those still event-free. It does not mean 50% higher cumulative event probability; interval reference is 1.

Measurement and model fit

Term / notation	Meaning	Range and numerical interpretation
α: alpha	In reliability analysis, α is Cronbach's internal-consistency coefficient. In hypothesis testing, α is a prespecified Type I error threshold, such as .05. These are different uses of the same symbol.Also written as: α · alpha	**Usually 0-1 in a coherent scale; estimates can be negative**For reliability alpha, .70 is a common rough screening benchmark, not universal adequacy. Very high values (around .95+) can suggest redundant items. Negative alpha calls for checking scoring and item covariance. Testing α has a different meaning.Tavakol & Dennick: alpha · UCLA: negative reliability estimates
λ: factor loading	In a factor model, λ links an indicator to a latent factor. Its scale depends on standardisation and identification. A standardised loading .80 implies .64 explained variance only under the stated simple-model assumptions.Also written as: λ · factor loading · factor loadings · loading · loadings · □□□□□ · □□□	**Scale/model dependent; loadings may have either sign**In a simple standardised one-factor model, |λ| ≤ 1 and λ = .70 implies .49 explained variance. Unstandardised or cross-loading coefficients need not share that bound. A simple-model loading above 1 can indicate an improper solution.
θ: latent trait or threshold	In an item-response model, θ usually denotes a person's latent trait level. In an ordinal regression equation, θ_j may denote category threshold j. The equation determines the interpretation.Also written as: θ	**Usually any real value on the latent/link scale**An IRT scale is often identified with mean 0 and SD 1; θ = +1 is then one latent SD above the reference mean, not a 1% gain. Ordered thresholds must follow their specified ordering.
h²: communality	The part of an indicator's variance accounted for by the common factors under the fitted factor model. The sum of squared loadings is sufficient only under the stated orthogonal-factor conditions.Also written as: h²	**0 ≤ h² ≤ 1 for an admissible standardised factor model**h² = .64 means common factors account for 64% of that indicator's variance. Values above 1 are a warning of an inadmissible/Heywood-type solution, not especially strong measurement.
φ: context-dependent association	For a 2×2 table, φ is an association effect size related to chi-square. In factor-model notation, φ may instead denote a factor correlation or covariance. Read the model context.Also written as: φ	**−1 to +1 for correlations; covariance has other units**For a 2×2 association or factor correlation, 0 indicates no association on that metric. If φ denotes covariance, its numerical range depends on the variables and is not fixed at ±1.

Term / notation	Meaning	Range and numerical interpretation
ICC: intraclass correlation	In a simple Gaussian random-intercept model, ICC=between-cluster variance/(between-cluster variance+residual variance). ICC=.20 means 20% of the modelled variance is between clusters. Other ICC definitions serve other designs.Also written as: ICC · intraclass correlation · □□□□□□	**0-1 for the simple variance-component definition**ICC = .20 means 20% of modelled variance is between clusters. Other sample ICC estimators can be negative. There is no universal cutoff for ignoring clustering or declaring rater agreement adequate.
CFI: comparative fit index	An incremental fit index comparing a specified factor/SEM model with a baseline model. Read it alongside residuals, other fit indices, uncertainty and model plausibility; a single cutoff does not validate a model.Also written as: CFI	**Usually 0-1 in bounded implementations**Closer to 1 indicates better relative fit. CFI near .95 is a widely cited guideline from specific ML simulations, not proof of validity or a universal rule for ordinal/other estimators.Hu & Bentler: fit-index guidelines · Xia & Yang: estimator-dependent fit
TLI: Tucker-Lewis index	An SEM incremental fit index comparing model fit with a baseline while accounting for model degrees of freedom. Interpret it with the model specification, other fit indices and residual evidence.Also written as: TLI	**Often near 0-1; may fall below 0 or exceed 1**Closer to 1 generally indicates better relative fit. Near .95 is a commonly cited guideline in particular ML conditions. TLI = 1.02 is possible; read the estimator, df and other diagnostics.Hu & Bentler: fit-index guidelines · Xia & Yang: estimator-dependent fit
RMSEA	Root mean square error of approximation: an SEM fit index addressing approximate discrepancy relative to degrees of freedom. Read its interval and the model's df; small values alone do not establish a sound model.Also written as: RMSEA	**RMSEA ≥ 0; no universal upper bound**Lower indicates less approximate misfit; around .06 or below is a cited ML guideline, not an automatic pass. Read its interval and df; values can behave poorly at low df or small samples.Hu & Bentler: fit-index guidelines · Xia & Yang: estimator-dependent fit
SRMR	Standardised root mean square residual: a summary of standardised discrepancies between observed and model-implied association structures. A small overall index can still conceal important local misfit.Also written as: SRMR	**SRMR ≥ 0**0 means no discrepancy in the summarised standardised residuals; around .08 or below is a cited fit guideline. A small average can hide local misfit; read jointly with other diagnostics.Hu & Bentler: fit-index guidelines
BIC: Bayesian information criterion	A relative model-comparison criterion balancing fit and parameter count. Lower BIC is preferred among comparable candidate models fitted to the same data. It is not a p-value or an absolute certificate of model quality.Also written as: BIC	**May be negative or positive**Lower is preferred among appropriately comparable models fitted to the same data. BIC = −20 can be better than −10. Zero is not a pass/fail reference; neither the sign nor BIC alone selects a defensible model.
AIC: Akaike information criterion	A relative model-comparison criterion balancing fit and complexity. Lower is preferred among appropriately comparable models using the same data. AIC and BIC impose different complexity penalties.Also written as: AIC	**May be negative or positive**Lower is preferred among appropriately comparable models using the same response/data and likelihood basis. AIC = 100 versus 104 favours 100 on this criterion; no absolute "good AIC" exists.

Mediation, moderation and latent groups

Term / notation	Meaning	Range and numerical interpretation
M: mediator	In mediation notation, M is the intervening variable in the proposed X→M→Y pathway. Calling it a mediator in an equation does not by itself establish a causal mechanism.Also written as: M	**Variable label: no inherent numerical range**The range belongs to the mediator's measurement or coding: for example 0/1 groups, a 1-5 score or unrestricted values. The letter itself has no cutoff.

Term / notation	Meaning	Range and numerical interpretation
W: moderator	In the interaction examples, W is the moderator. The X-Y slope is evaluated at stated values of W. Different coding or centring changes which conditional slope is the main coefficient.Also written as: W	**Variable label: no inherent numerical range**The range belongs to the moderator's measurement or coding: for example 0/1 groups, a 1-5 score or unrestricted values. The letter itself has no cutoff.
a: context-dependent coefficient	In mediation, a labels the X→M path coefficient. In a 2PL item-response model, a is item discrimination, describing the steepness of the response curve. Read the model before interpreting the symbol.Also written as: a	**Usually signed; interpretation depends on the model**A mediation path a can have either sign, with 0 as the no-association reference. IRT discrimination is commonly positive for correctly keyed items; constraints and direction conventions must be checked.
ab: indirect component	For the specified linear mediation model without an X×M interaction, multiply a (X→M) by b (M→Y conditional on X). This gives the indirect component ab; its uncertainty needs an appropriate interval method.Also written as: ab	**Signed; no universal numerical bounds**The indirect product can be positive or negative. In the specified linear model, an appropriate interval excluding 0 supports a nonzero indirect component; it does not establish a causal mechanism.
c′: direct path coefficient	In the stated mediation model, c′ labels the X→Y coefficient conditional on mediator M and the other included variables. This statistical direct path requires additional assumptions for a causal interpretation.Also written as: c′	**Signed; no universal numerical bounds**0 is the no-direct-association reference on the specified coefficient scale, conditional on M and other included variables. A nonsignificant c′ does not establish "full mediation".
π: mixture proportion	In a mixture model, π_k is the modelled proportion of cases in profile or class k. Proportions across all components sum to 1. A class proportion differs from an individual's posterior membership probability.Also written as: π_k · π□ · π□ · π	**0 ≤ πk ≤ 1; class proportions sum to 1**πk = .30 means a modelled 30% class/profile proportion. It is not the probability that one particular learner belongs to that class; no fixed minimum class proportion validates a solution.
Entropy in latent grouping	A summary of posterior classification separation for one fitted class/profile solution. Higher separation does not by itself determine the number of groups or show that groups are natural types.Also written as: entropy · Entropy	**0-1 for the normalised LPA/LCA index**Closer to 1 means clearer classification; .80 is often used as a rough guide to separation. It is not a p-value, a percentage correctly classified, or a rule for choosing the class count. Other entropy definitions have different ranges.Mplus: entropy and classification

Estimation, prediction and research synthesis

Term / notation	Meaning	Range and numerical interpretation
OLS: ordinary least squares	Fits a linear model by minimising the sum of squared residuals. The estimator, inferential standard errors and independence assumptions are separate choices that should be specified.Also written as: OLS	**An estimation method, not a bounded statistic**OLS does not make all coefficients positive or impose a significance cutoff. Coefficient ranges follow the specified model.
ML: maximum likelihood	Chooses parameters that maximise the probability density/mass of the observed data under the specified model. ML is an estimation principle, rather than a substantive analysis method or guarantee that the model is correct.Also written as: ML · maximum likelihood · □□□□	**An estimation method, not a bounded statistic**A likelihood is non-negative, but a likelihood density can exceed 1 and a log likelihood can have either sign. ML itself is a fitting principle, not a number to compare with .05.

Term / notation	Meaning	Range and numerical interpretation
FIML: full-information ML	Uses each case's observed measurements in a model-based likelihood for incomplete data. Validity relies on the model and missingness assumptions, such as MAR with appropriate observed variables. It does not mean every missing value is known or filled in.Also written as: FIML	**An estimation method, not a bounded statistic**FIML has no numerical pass mark and is not the percentage of complete data. Interpret the resulting parameters and the missing-data assumptions.
Bootstrap	Repeatedly resamples the data under a justified scheme and recomputes a statistic to approximate its sampling uncertainty. Resample people or clusters when the design requires it; resampling individual rows may break dependence.Also written as: bootstrap · resampling · □□□	**An estimation method, not a bounded statistic**The number of replicates is a positive integer, but there is no universal bootstrap cutoff. The statistic's range and the interval type determine interpretation.
Posterior probability	A probability updated using data under a specified model. In LPA/LCA it can express a case's uncertain profile/class membership. A highest-probability assignment does not make membership certain.Also written as: posterior probabilities · posterior probability · □□□□	**0-1 for a posterior probability**A class probability of .80 means 80% membership probability under the fitted model, not certainty. Posterior densities, unlike probabilities, may exceed 1; posterior parameters have their own domains.
Credible interval	A Bayesian posterior interval containing a stated probability mass under the model and prior. Its probability interpretation differs from the repeated-sampling coverage of a frequentist confidence interval.Also written as: credible interval · credible intervals · □□□□	**Endpoints follow the parameter domain**A 95% credible interval contains 95% posterior probability under the model and prior. Compare signed effects with 0 and ratios with 1, but do not equate posterior coverage with frequentist confidence coverage.
Prediction interval	An interval for a future observation or effect under a specified model. It includes relevant future variability, and answers a different question from a CI for a mean or coefficient.Also written as: prediction interval · prediction intervals · □□□□	**Endpoints follow the predicted outcome/effect**A 95% prediction interval targets future variation under the specified model. It can span 0 even when the CI for the mean effect does not; there is no universal acceptable width.
RMSE: prediction error	Root mean squared prediction error: square root of the mean of (observed−predicted)². It is in outcome units; lower is better on the same evaluation data and task. Training RMSE is not an independent validation result.Also written as: RMSE	**RMSE ≥ 0**0 = exact predictions on the evaluated data. Lower is better for the same task and evaluation data; RMSE = 2 is two outcome units of root-mean-square error, not 2%.
τ^2: between-study variance	In a random-effects meta-analysis, τ^2 describes heterogeneity in the underlying study effects, beyond their sampling variances. It is a variance, expressed in squared effect-size units.Also written as: τ^2	**$\tau^2 \geq 0$ in a constrained random-effects model**0 indicates an estimated absence of between-study variance, not proof of homogeneous true effects. τ^2 = .04 corresponds to τ = .20 in the effect-size units; no universal small/large cutoff applies.

Make quantitative analysis and results transparent

Alongside JARS-Qual in Topic 7, use JARS-Quant for this lesson. This classroom companion condenses reporting prompts; consult the complete table and target journal. A tick means you can locate the reported evidence, not that the analysis is valid.

APA JARS-Quant · Table 1

[ ] **Questions, hypotheses and status**Name the target quantities; distinguish primary, secondary and exploratory analyses.

[ ] **Design, sampling and setting**Describe recruitment, inclusion/exclusion, selection, settings and dates.

[ ] **Sample size, power and precision**Explain intended and achieved sample sizes and their rationale; report participant flow.

[ ] **Ethics and data collection**Identify ethical review, consent and procedures relevant to the study.

[ ] **Measures and scoring**Define variables, covariates, units, item coding and composite-score rules.

[ ] **Measurement evidence**Report relevant reliability in the current sample and validity/comparability evidence.

[ ] **Missingness and exclusions**Report amounts/patterns, handling, outlier decisions, exclusions and final analysis n.

[ ] **Model, estimand and estimation**Specify variables, contrasts, estimators, dependence, weighting and transformations.

[ ] **Assumptions and diagnostics**Explain relevant checks, estimation failures and responses to violations.

[ ] **Descriptives and all relevant results**Provide sample summaries and planned results, including non-significant findings.

[ ] **Magnitude and uncertainty**Report estimates, units, intervals, effect sizes and inferential statistics as applicable.

[ ] **Multiplicity and sensitivity**Disclose multiple tests, adjustments, alternative models and unplanned changes.

[ ] **Tables, figures and reproducibility**Label units, n, intervals and notes; identify software, versions and access to data/materials/code where appropriate.

[ ] **Interpretation and limitations**Discuss precision, generalisability, alternatives and design-based limits on causal claims.

Thank you

Thank you for your questions and participation. Start with the research question, show the data clearly and explain the uncertainty.

Download the English lesson PDF to revisit the basics, method-map links, glossary and JARS-Quant checklist. The full advanced method explanations remain available through the interactive map.

DEDC02 course materials